(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-104 **Subject: Applied Chemistry**

[Batch 2021 Onwards]

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q.1. **Attempt any three.** (3×5=15)

(a) On burning 0.83 g of a solid fuel in a bomb calorimeter, the temperature of 3500 g of water increased from 26.5°C to 29.2°C. Water equivalent of calorimeter and latent heat of steam are 385.0 g and 587.0 cal/g respectively. If the fuel contains 0.7% hydrogen, calculate its gross and net calorific value.

(b) State Gibb's Phase rule. With the help of suitable example explained the terms:
i) Components ii) Degree of freedom iii) Phase

(c) What are conducting polymers? Illustrate with an example.

(d) Calculate the quantity of lime and soda required to soften 20,000 liters of water containing the following salts:
$CaCO_3$ =10.0 mg/L, $MgCO_3$ =8.4 mg/L, $CaCl_2$ =11.1 mg/L, $MgSO_4$ =6.0 mg/L, SiO_2 =1.2 mg/L
Assuming the purity of lime as 90% and soda as 95%.

(e) Discuss sacrificial anodic protection. What is the condition for a metal to act as a sacrificial anode to iron?

UNIT-I

Q.2. (a) Define carbonization of coal.Explain Otto-Hoffman oven method of carbonization and recovery of by-products with diagram. What are its advantages over earlier methods? (10)

(b) What do you understand by the term knocking in IC engine? Explain its significance with chemical constituents of fuels. (5)

OR

Q.3. (a) What are advantages of catalytic cracking of petroleum? Explain fixed bed catalytic method of cracking. Give its mechanism. (10)

(b) Discuss the relative merits and demerits of solid, liquid and gaseous fuels. (5)

UNIT-II

Q.4. (a) What do you mean by congruent melting point? Discuss the phase diagram of a two-component system with the formation of a compound having congruent melting point. (6)

(b) Give the preparation properties and uses of the following (3x3=9)
i) Phenol formaldehyde resin
ii) LDPE
iii) Nylon

P.T.O.